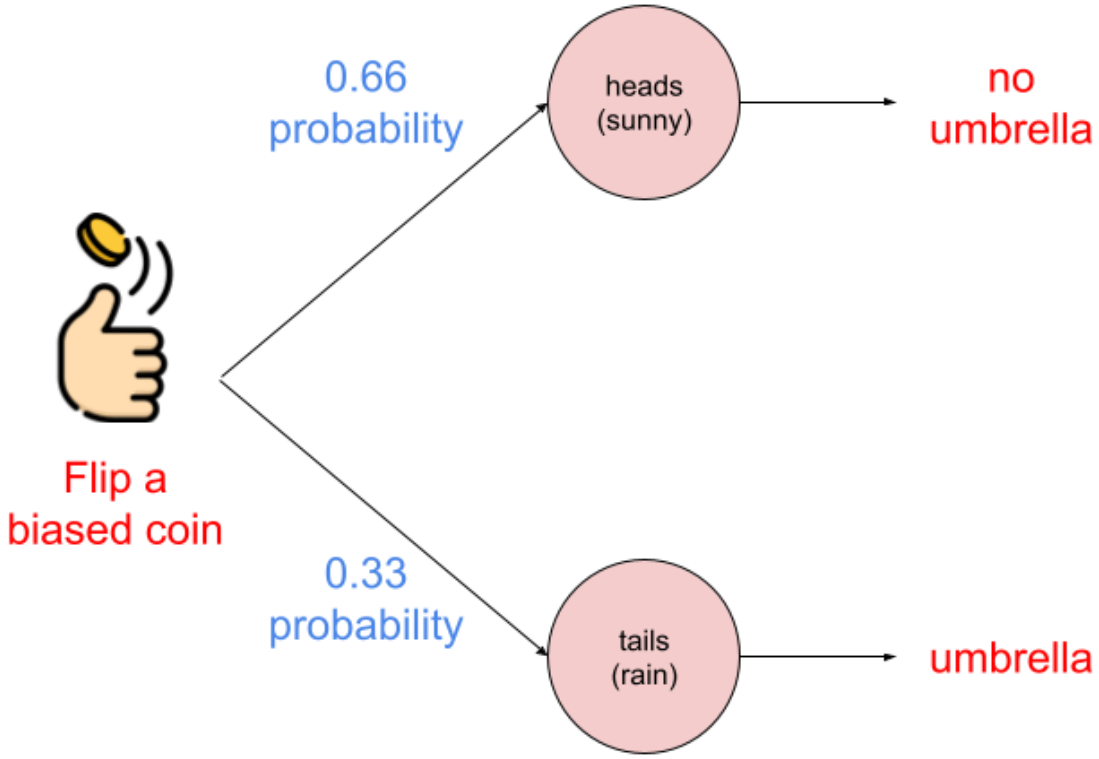


Figure 5.11 The Thompson sampling policy as a decision tree. We flip a biased coin to obtain a sample from the probability distribution of sunny weather and decide whether to bring an umbrella based on this sample.

While this might seem like an arbitrary method of decision making at first, TS is particularly effective in both MAB and BayesOpt. First of all, given a probability distribution representing a quantity of interest, a sample of that distribution is a possible realization of that quantity, so that sample may be used as a representation of the distribution. In optimization under uncertainty problems, TS offers the same benefit as UCB in that sampling from a probability distribution is often easy to do. Just like how UCB's optimistic estimates of the reward rates may be generated in an efficient manner, sampling may be done just as efficiently.

Further, by using a random sample, we are still balancing between exploration and exploitation. To see this, let's consider how TS works in BayesOpt. From a GP trained on the current observed data, we draw *one* sample. Remember from Chapter 3 that a sample from a GP is a function representing a particular realization of the objective function according to our GP belief. However, unlike the true objective function which is unknown in regions without observed data, a sample drawn from a GP is entirely known. This means that we can find the location at which this sample is maximized.

Figure 5.12 shows the GP we trained on the Forrester function and three samples drawn from it as dashed lines. Further, along each of the samples, we use a diamond to indicate the location at

©Manning Publications Co. To comment go to liveBook
https://livebook.manning.com/#!/book/bayesian-optimization-in-action/discussion

Licensed to ashwin sampath <ashwin.manning@gmail.com>